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Climate discourses of petroleum corporations in China and the United States: A comparative stakeholder analysis

Jieyun Feng [a], Wenze Lu [b,*], Ming Liu [c], Wenqing Yu [a]

[a] Department of Business English, School of International Studies, University of International Business and Economics, Beijing, 100029, China
[b] School of Nursing, Faculty of Health and Social Sciences, The Hong Kong Polytechnic University, HKSAR, 999077, China
[c] Department of Chinese and Bilingual Studies, Faculty of Humanities, The Hong Kong Polytechnic University, HKSAR, 999077, China

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ABSTRACT

Although corporate climate discourse has been extensively studied, research has primarily focused on self-legitimisation and self-promotional strategies. How corporations engage with key stakeholders and balance competing priorities remains underexplored. By integrating stakeholder theory with corporate climate discourse, this study explores how American and Chinese petroleum corporations frame their climate discourses in relation to two key global and national stakeholders: the Paris Agreement and their respective national governments. Using a mixed-methods approach, this study conducts a quantitative thematic analysis of the Paris Agreement and national climate policies, followed by a qualitative thematic analysis of climate discourse from six leading petroleum corporations—Marathon Oil, ExxonMobil, and Chevron (U.S.) and CNPC, Sinopec, and CNOOC (China). Findings indicate that the Paris Agreement prioritises global engagement, development capacity and a formal, obligatory commitment to directives. China's policy discourse presents the government as a central orchestrator, actively guiding national policy within a global framework. Meanwhile, U.S. policy discourse focuses on addressing immediate climate crises and promoting environmental justice while emphasising the roles of various agencies. In response to these two key stakeholders, Chinese corporations align their discourse closely with the Paris Agreement, centring on two themes: 'transition to sustainable energy management' and 'policy-driven climate strategy'. American corporations take a more delicate approach, balancing global commitment with business operations while demonstrating 'selective policy support and active lobbying' and 'advocacy for market-driven carbon pricing'. This study concludes with a discussion of its academic contributions and practical implications for climate regulators, corporate communication students and practitioners.
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* Corresponding author.
E-mail addresses: fengjieyun88@126.com (J. Feng), chriswenze.lu@polyu.edu.hk (W. Lu), ming1.liu@polyu.edu.hk (M. Liu), yuwq990915@126.com (W. Yu).

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1. Introduction

As climate change emerges as a pressing global issue (Machin & Liu, 2024), petroleum corporations are under growing social pressure to reduce their carbon footprint. Although petroleum remains economically indispensable for global transportation, power generation and industrial development, its environmental impact is becoming increasingly evident (Intergovernmental Panel on Climate Change, 2018). Therefore, petroleum companies are under increasing scrutiny and criticism from international organisations and environmental advocates (Wang et al., 2022; Wright & Nyberg, 2012). This dual role—economic necessity and environmental harm—positions petroleum corporations as central players in global climate change discussions.

Given the wide-ranging global impact of climate change, it is perhaps unsurprising that a need for robust international cooperation has been identified (the United Nations Paris Agreement, 2015). In this context, China and the United States (U.S.) play particularly crucial roles as the world's two largest carbon emitters (Guo et al., 2023), thus making their policies and actions pivotal in addressing climate challenges. Both nations have committed to ambitious carbon neutrality goals—China by 2060 and the U.S. by 2050—and have introduced measures to reduce reliance on fossil fuels (Wu et al., 2022). Petroleum corporations in China and the U.S. have been urged by international organisations and policymakers to reform their corporate practices and align with decarbonisation efforts (Bhatia, 2013; Wang et al., 2022). How these corporations navigate this transition and engage with key stakeholders will be crucial in helping China and the U.S. meet their climate commitments and influence broader global climate initiatives.

Given this complexity, examining the climate discourse of Chinese and American petroleum corporations is essential. This study analyses how these corporations respond to two critical stakeholders: the Paris Agreement and their respective national governments. By contributing to existing research on corporate climate discourse, the study offers valuable insights for climate policymakers, ESP students, and corporate climate communication practitioners.

This study continues with a review of recent literature on corporate climate discourse, with particular reference to stakeholder theory and stakeholder discourse. Subsequently, an integrated theory of corporate climate discourse and its stakeholders is proposed, along with three specific research questions. Following a discussion of our research methodology, this study presents a detailed analysis of the results. It then closes with a critical evaluation of what these results may mean and a consideration of their broader implications.

2. Literature review

2.1. Stakeholder theory and discourse

Stakeholder theory, introduced by Freeman (1984), represents a paradigm shift from traditional shareholder primacy. The theory posits that a corporation's stakeholders extend beyond shareholders to include individuals and groups affected by its actions, decisions and policies—such as investors, employees, customers, suppliers, the local community and regulators. This theory emphasises that all stakeholders, given their investments and shared risks, have legitimate claims on the corporation's operations and growth. Hence, corporations must account for diverse stakeholder interests to achieve sustainable growth and long-term viability (Schaltegger et al., 2019).

Building on stakeholder theory, the concept of stakeholder discourse emerges, encompassing complex language use and communication among those who influence or are influenced by corporations (Mitchell et al., 1997). Moreover, analysing stakeholder discourse is essential for understanding their perspectives and concerns in corporate decision-making while fostering transparency and inclusivity in business operations (Archetti, 2000). Stakeholder discourse is rarely examined in isolation; instead, it is often analysed along with the dynamic interactions between corporations and their stakeholders to reveal how they connect, communicate, reconcile, compromise and compete. For example, Gottschick (2015) employed a discourse analysis framework based on four distinct conceptualisations of climate uncertainty, applying it to interpret interviews with diverse stakeholder groups within a local governance adaptation network. Similarly, Feng et al. (2017, 2018) examined the discourses of five different stakeholders—UNESCO, national government regulators, international tourists, domestic tourists and business operators, thus identifying competing images and a continuum between heritage preservation and tourism growth.

2.2. Research on corporate climate discourse

Generally, corporate climate discourse is considered a part of strategic or impression management (Ahmad & Hossain, 2019; Van Halderen, Bhatt, Berens, Brown, & van Riel, 2016), designed to position corporations as active contributors to climate action while meeting the expectations of diverse stakeholders. A corporation's stance on climate change should be shaped by shareholders, who primarily seek financial returns, and a broad range of stakeholders, including government regulators (Reid & Toffel, 2009), executives (He et al., 2021) and environmentally conscious consumers (Megura & Gunderson, 2022). It is important to recognise that different stakeholder groups—such as investors and the general public—may have conflicting expectations, and these tensions can be reflected in corporate discourse (Livesey, 2002).

Studies on corporate climate discourse often adopt functional and critical perspectives, revealing how major energy corporations employ self-legitimisation and promotional strategies to shape their stance on climate change amid increasing

social pressure (Dahl, 2022; Megura & Gunderson, 2022; Wang & Liu, 2024). These studies typically involve cross-time and -country dimensions. In terms of both dimensions, Fuoli and Beelitz (2024) examined the sustainability reports of the world's largest greenhouse gas (GHG) emitters from 2011 to 2020. Their findings show an increased emphasis on climate change following the Paris Agreement but little consensus among major carbon emitters on achieving carbon neutrality—one of the agreement's key objectives. Furthermore, companies in middle-income countries tend to focus less on climate change compared with those in high-income countries. Regarding cross-national comparisons, Bhatia (2012) conducted a genre and critical discourse analysis of corporate social responsibility (CSR) reports from American and Chinese oil companies. The findings suggest that CSR reports from both countries employed three interdiscourses, namely, self-promotion, goodwill and self-justification. Specifically, they used self-justification discourse to attribute company actions to external constraints. From a diachronic perspective, Jaworska's (2018) study examined how major oil companies framed climate issues in CSR reports and how their discourse evolved between 2000 and 2013. The findings suggest that in the mid-2000s, climate change was perceived as a manageable issue with potential solutions. However, in recent years, climate change has been increasingly framed as a risk—portrayed as an unpredictable and uncontrollable force. This shift reflects a distancing strategy, repositioning climate change as a future concern and attributing responsibility to other stakeholders.

Several notable studies have examined how corporations frame climate discourse in terms of stakeholders, including the media. Kapranov (2017) conducted a qualitative study using conceptual metaphor analysis to explore the Shell Group's corporate climate image and its representation in leading British financial newspapers, *The Economist* and *The Financial Times*. It compared and discussed the discrepancies between Shell's positive climate image and its more mixed portrayal by these media outlets. Moreover, Fuoli and Beelitz (2024) explicitly connected corporate discursive practices to the landmark Paris Agreement. However, since their study did not focus on systematically analysing specific stakeholder groups, it did not explore the full linguistic patterns of the Paris Agreement or its broader engagement with corporate climate discourse.

In summary, two critical research gaps have been identified in the existing literature. First, while stakeholder discourse has been extensively studied, research on corporate climate discourse has primarily focused on self-legitimisation and promotional strategies. The discursive dynamics between corporations and various stakeholders remain largely unexplored, particularly in relation to two key actors: the global climate regulator (i.e. the Paris Agreement) and national government regulators. Given their significant influence in shaping corporate climate discourse, an in-depth academic investigation is warranted. The Paris Agreement, as a legally binding framework for signatory nations, exerts significant global influence, while national governments formulate concrete climate policies and regulatory guidelines for corporations. Second, most studies (e.g. Bhatia, 2012; Fuoli & Beelitz, 2024; Jaworska, 2018) have relied on specialised reports, such as sustainability and CSR reports, as primary data sources for analysing corporate climate discourse. Alternative platforms, such as corporate webpages and social media posts, provide diverse and dynamic insights into corporate messaging, enabling more regular and interactive communication with stakeholders. A discourse analysis that draws data from multiple sources offers a more comprehensive understanding of corporate climate communication.

3. Theory and research questions

This study applies stakeholder discourse theory to examine corporate climate communication (Archetti, 2000; Gottschick, 2015; Mitchell et al., 1997). This theory offers a holistic and dynamic framework, conceptualising discourse as the continuous interaction between corporations and their stakeholders, wherein they engage, negotiate and reconcile competing interests. 'Discourse' refers to the concrete language used by different stakeholders within a specific communicative context (Feng, 2017, 2018). Through nuanced linguistic analysis, this approach provides an in-depth understanding of the complex relationship between corporate climate discourse and various stakeholders, thereby uncovering underlying motivations and multiple influences. Adopting a stakeholder-focused perspective, this study conducts a comparative analysis of the climate discourse of petroleum corporations in China and the U.S. Hence, this study investigates the following three research questions:

Research Question 1. What are the climate change stances of the Paris Agreement and the national governments of China and the U.S.?

Research Question 2. What stances do Chinese and American petroleum corporations take regarding the Paris Agreement?

Research Question 3. What stances do Chinese and American petroleum corporations take in relation to their respective national governments?

4. Research methodology

4.1. Data collection for two stakeholders

The Paris Agreement and the national governments of China and the U.S. were selected as key stakeholders in corporate climate discourse, as they represent influential global and national regulators in climate policy.

The Paris Agreement was selected as a data source to represent the stance of the global regulator, the United Nations. This global climate pact reflects the collective commitment of the international community to combat climate change. Adopted on 15 December 2015, and entering into force on 4 November 2016, it was endorsed by 195 signatory nations—including China, the U.S. and the European Union—during the 21st Session of the Conference of the Parties (COP21). Emerging from extensive multilateral negotiations, the agreement exemplifies the United Nations' dedication to fostering international cooperation on environmental issues (United Nations, n.d.). The document comprises a total of 7,587 tokens.

The Chinese and American governments develop and enforce domestic climate policies that set compliance and operational standards, exerting a significant influence on corporate climate discourse. As the world's two largest GHG emitters, China and the U.S. play a pivotal role not only in shaping corporate climate initiatives but also in global climate governance.

A total of eight official documents—four from China and four from the U.S.—focusing on national climate policies were collected from 2021 to 2022, during the post–Paris Agreement period. To ensure comparability between the two nations, this time frame was chosen when both governments were actively enforcing the agreement. While China upheld the agreement throughout these years, the U.S. formally withdrew from the global climate pact during President Donald Trump's administration. Soon after President Joe Biden took office, the U.S. rejoined the Paris Agreement on 19 February 2021 and has since continued to enforce the pact. The documents were sourced from the official websites of the State Council of the People's Republic of China and the White House. The corpus of Chinese policies consists of four key documents related to energy and climate, totalling 159,281 tokens. For the U.S. side, data were collected during President Joe Biden's tenure. The U.S. policy corpus comprises four executive orders, totalling 120,255 tokens. Table 1 presents the data collected from the two stakeholders: the Paris Agreement and the national governments of the U.S. and China.

Table 1
Summary of Collected Data from Two Key Stakeholders.

Two Stakeholders	Data sources	Data size
Paris agreement	Paris agreement in 2015	7,587 tokens
Chinese national government	1. Energy work guidelines in 2021 (2021年能源工作指导意见); 2. Energy work guidelines in 2022 (2022年能源工作指导意见); 3. Guidelines on improving institutional mechanism, policies and measures for green and low-carbon energy transition in 2022 (2022年关于完善能源绿色低碳转型体制机制和政策措施的意见); 4. The state Council's guideline on achieving the goals of carbon peak and carbon neutrality with full and faithful implementation of the new development philosophy in 2021 (2021年国务院关于完整准确全面贯彻新发展理念做好碳达峰碳中和工作的意见)	159,281 tokens
American national government	1. Climate-related financial risk in 2021; 2. Catalyzing clean energy industries and jobs through federal sustainability in 2021; 3. Protecting public health and the environment and restoring science to tackle the climate crisis in 2021; 4. Tackling the climate crisis at home and abroad in 2021	120,255 tokens

4.2. Data collection for petroleum corporations

Data were collected from six leading petroleum corporations based in China and the U.S., all of which were listed in the Fortune 100 for 2022. The selection criteria focused on the top three oil corporations from each country: China National Petroleum Corporation (CNPC), China Petroleum & Chemical Corporation (Sinopec) and China National Offshore Oil Corporation (CNOOC) from China, along with ExxonMobil, Marathon Oil and Chevron from the U.S.

Corporate discourse on climate change is multifaceted. In March and April 2022, we collected official webpages from these six corporations dedicated to climate change. The content varied, encompassing 16 pages that addressed themes such as climate management and emissions control, along with additional linked pages within the main climate change sections. Furthermore, we compiled 12 independent climate or sustainability reports from Chinese and American corporations published in 2021 and 2022. Relevant posts were further gathered from two social media platforms—Twitter in the U.S. and Weibo in China. In total, 52 posts from 2021 to 2022 related to climate change and GHG emissions were selected. The relatively small number of posts is due to strict selection criteria, as irrelevant or off-topic content was filtered out to ensure a focused analysis of climate change and emissions. Refer to Table 2 for the collected data on petroleum corporations.

Table 2
Summary of Collected Corporate Data from Six Leading Petroleum Corporations in China and the U.S.

Corporations	Corporate Webpages (2022)	Corporate Climate Reports (2021–2022)	Social Media Posts (2021–2022)
CNOOC	CNOOC environmental protection	2021 CNOOC environmental, social and governance report 2022 CNOOC environmental, social and governance report	0 posts
CNPC	CNPC climate change CNPC tackling climate change	2021 CNPC corporate social responsibility report 2022 CNPC corporate social responsibility report	7 posts
Sinopec	Sinopec climate change Sinopec tackling climate change	2021 sinopec corporate social responsibility report 2022 sinopec corporate social responsibility report	6 posts
ExxonMobil	ExxonMobil advancing climate solutions ExxonMobil blue hydrogen climate solutions ExxonMobil outlook for energy	ExxonMobil advancing climate solutions 2022 progress report ExxonMobil 2021 energy & carbon summary	23 posts
Marathon oil	Marathon oil climate change Marathon oil emissions management Marathon oil political contributions & lobbying	2021 marathon climate report 2022 marathon sustainability report	4 posts
Chevron oil	Chevron climate policy Chevron supports well-designed climate policy Chevron well-designed methane policy Chevron supports innovation to scale climate solutions	2021 chevron climate change resilience report Chevron sustainability report 2022	12 posts

4.3. Research methods

This study adopted a comparative discourse analysis approach to examine the stances of two key stakeholders and how petroleum corporations responded to them. We incorporated qualitative and quantitative research methods to address different data types.

4.3.1. Quantitative thematic analysis for two stakeholders' discourse

To examine the themes of two key stakeholders—the discourse of the Paris Agreement and the climate policies of the American and Chinese national governments—we used the corpus analysis tool Wmatrix 3.0 for semantic domain analysis. This tool was selected for its advanced capabilities in automatically analysing semantic domains and distilling the overarching themes of the discourse (Liu & Huang, 2022). The corpus tool is equipped with the CLAWS tagging system and the UCREL Semantic Analysis System, thus enabling automatic tagging of parts of speech and semantic categories. This tool facilitates corpus comparison against an embedded reference corpus, automatically generating semantic domain distribution tables (Rayson, 2008). The generalised semantic domains can be grouped into broader categories, aiding in the identification of overarching themes. Furthermore, by using the keyword list and Key Word in Context concordance indices generated by Wmatrix, the contextual features of keywords and semantic domains can be revealed.

Using the corpus tool Wmatrix 3.0, we identified the top 10 key semantic domains across three corpora, namely, the official document of the Paris Agreement and the climate policies of the American and Chinese national governments. These corpora were compared against the British National Corpus Sampler written as a reference corpus. After analysing the results, the authors discussed the 10 categories, identified broader patterns and extracted prominent themes. Moreover, we qualitatively examined the themes, using illustrative examples to fully contextualise the keywords within concrete discourse.

4.3.2. Qualitative thematic analysis of corporate climate discourse

We examined corporate climate discourse from China and the U.S. using qualitative thematic analysis. This approach was necessary due to the diverse sources of data and the relatively limited size of text-based material, which was insufficient for forming a corpus suitable for quantitative analysis (Dodgson, 2017). To be specific, this study followed the six-step methodology outlined by Braun and Clarke (2006), including familiarisation, coding, searching, reviewing, defining and writing up. The initial stage, familiarisation, involves comprehensive, critical and active reading of the data to extract meaningful insights. During the coding phase, the authors identified keywords and phrases relevant to the research questions and categorised them as codes. Rather than using a predefined coding scheme, this study adopted an inductive approach, allowing patterns to emerge naturally from the data. This led to the searching for themes stage, where overarching patterns were defined as themes—coherent and meaningful structures within the discourse. Subsequent stages included reviewing themes to ensure consistency with the collected data and defining and naming themes, where each theme was meticulously labelled to accurately reflect its essence. The final stage, writing up, detailed the significance of each theme through data excerpts, providing a comprehensive analysis of corporate discourse on climate issues.

In analysing corporate climate discourse, we examined how the datasets related to the Paris Agreement and their respective national governments, assessing the extent to which they aligned with key themes in the discourse of these two stakeholders. Throughout the analytical process, the lead authors—with much expertise in discourse and linguistic research—worked closely to discuss, cross-check and reach a consensus on theme identification.

5. Results

5.1. Stances of the Paris Agreement on climate change

As presented in Table 3, the semantic domains identified in the Paris Agreement—sorted by log-likelihood (LL) values—include S1.1.3+ (Participating), A6.1+ (Comparing: Similar), Q4.2 (The Media: Newspapers), S8+ (Helping), A2.1+ (Change), A1.8+ (Inclusion), T1.1.3 (Time: Future), M7 (Places), W4 (Weather) and X2.2+ (Knowledgeable).

Table 3
Key Semantic Domains and Keyword in the Paris Agreement.

Rank	Tagset[a]	LL[b]	Semantic Domain	Keywords
1	S1.1.3+	1,612.66	Participating	Parties, conference, meeting
2	A6.1+	410.48	Comparing: Similar	Agreement
3	Q4.2	386.54	The media: Newspapers	Article, articles
4	S8+	262.92	Helping	Support, promote, guidance
5	A2.1+	186.32	Change	Adaptation, change, development
6	A1.8+	160.12	Inclusion	Including, integration, taking-into-account
7	T1.1.3	133.75	Time: Future	Shall, will
8	M7	124.54	Places	Developing country, nationally
9	W4	121.41	Weather	Climate, weather
10	X2.2+	101.44	Knowledgeable	Information, recognising

[a] Tagset is a collection of assignments that represent semantic categories.
[b] LL represents log-likelihood, with higher LL values indicating greater statistical significance.

These domains are categorised into three thematic areas, namely, 'engagement', 'development' and 'prescriptive'. Within the 'engagement' theme, key domains include S1.1.3+ (Participating), S8+ (Helping) and M7 (Places). Keywords such as Parties', 'conference', 'support' and 'developing countries' highlight the cooperative nature of the Paris Agreement, consistently emphasising the necessity of joint action and collaboration. Example 1 underscores the significance of international support in climate adaptation, particularly for 'developing country Parties', which often face resource constraints and significant adaptation challenges. Example 2 offers guidance on engagement and advocacy for climate action, emphasising the importance of 'support the synthesis of relevant information and knowledge' and the 'provision of technical support and guidance'.

(1) *Parties recognize the importance of support for and* ***international cooperation*** *on adaptation efforts and the importance of taking into account the needs of* ***developing country Parties.***
(2) *Strengthening institutional arrangements to* ***support the synthesis of relevant information and knowledge,*** *and the* ***provision of technical support and guidance*** *to Parties.*

The 'development' theme, focusing particularly on capacity development, includes the domains A2.1+ (Change) and W4 (Weather). Keywords such as 'adaptation', 'change', 'climate' and 'weather' highlight the focus on environmental dynamics and the necessity of adapting to evolving climatic conditions, exemplified as follows:

(3) *Parties hereby establish the global goal on adaptation of enhancing* ***adaptive capacity****, strengthening resilience and reducing vulnerability to climate change, with a view to contributing to sustainable development and ensuring* ***an adequate adaptation response*** *in the context of the temperature goal.*
(4) *Increasing the* ***ability to adapt*** *to the adverse impacts of climate change and foster climate resilience and low greenhouse gas emissions development.*

The 'prescriptive' theme highlights the directive nature of the Paris Agreement, emphasising the vital role of knowledge and information dissemination in shaping future actions. This theme includes domains such as T1.1.3 (Time: Future) and X2.2+ (Knowledgeable). A key grammatical feature of the document is the frequent use of the modal verb 'shall', which signifies formal and obligatory commitments expected of the Parties, thereby establishing definitive and mandatory directives. The following examples illustrate this.

(5) *Each Party **shall** prepare, communicate and maintain successive **nationally determined contributions** that it intends to achieve.*
(6) *Developed country Parties **shall biennially communicate indicative quantitative and qualitative information** …*

5.2. Stances of the Chinese and American governments on climate change

5.2.1. Stances of the Chinese government on climate change

As presented in Table 4, Chinese climate policy discourse is characterised by 10 prominent semantic domains: X5.2+ (Interested/excited/energetic), S8+ (Helping), A2.1+ (Change), W5 (Green issues), I4 (Industry), Y1 (Science and technology in general), S1.2.5+ (Tough/strong), O1.3 (Substances and materials: Gas), A5.1+ (Evaluation: Good) and M7 (Places).

Table 4
Key Semantic Domains and Keywords in the Chinese National Government Climate Policies.

Rank	Tagset[a]	LL[b]	Semantic Domain	Keywords
1	X5.2+	2,435.87	Interested/excited/energetic	Energy
2	S8+	1,069.42	Helping	Promote, support, cooperation
3	A2.1+	906.88	Change	Transformation, development, reform
4	W5	737.77	Green issues	Environmental, energy conservation
5	I4	596.79	Industry	Industry, industrial
6	Y1	570.11	Science and technology in general	Technology, technological
7	S1.2.5+	565.07	Tough/strong	Strengthen
8	O1.3	557.54	Substances and materials: Gas	Gas, carbon dioxide
9	A5.1+	492.39	Evaluation: Good	Improve, enhance
10	M7	356.93	Places	National, international

[a] Tagset is a collection of assignments that represent semantic categories.
[b] LL represents log-likelihood, with higher LL values indicating greater statistical significance.

These domains can be categorised into three main themes, namely, 'energy management', 'green transformation with technology' and 'steering national policy within a global framework'. 'Energy management' is the most emphasised theme, highlighted by the domain X5.2+ (Interested/excited/energetic), which prominently features the keyword 'energy'. This domain reflects a strong focus on energy management in China's governmental discourse on climate policy, suggesting a strategic shift towards energy conservation and the use of renewable resources—as alternative energy sources and foundational elements for emerging industries and economic models. The following example illustrates this.

(7) *大力发展**非化石能源**, 着力培育能源新产业新模式, 持续优化能源结构 … (actively promote the development of non-fossil energy, prioritise the cultivation of new industries and innovative energy models and continue optimising the energy structure) …* (Energy Working Guidance in 2022)

The theme of 'green transformation with technologies' is represented by the domains A2.1+ (Change) and Y1 (Science and technology in general). Keywords such as 'development', 'transformation', 'reform', 'technology', 'industrial' and 'strengthen' indicate a shift towards a low-carbon economy, driven by technological advancements. This highlights an intent to leverage technology not only for economic growth but also as a crucial component of China's broader climate strategy. The following example illustrates this.

(8) *统筹国内国际能源资源, 推广**先进绿色低碳技术和经验** … (We will integrate domestic and international energy resources while **advancing green and low-carbon technologies and best practices**) …* (the State Council's Guideline on Achieving the Goals of Carbon Peak and Carbon Neutrality with Full and Faithful Implementation of the New Development Philosophy in 2021)

The theme of 'steering national policies within a global framework' is shaped by the domains M7 (Place) and S8+ (Helping). Keywords such as 'promote', 'support', 'national' and 'international' highlight the policy direction set by the Chinese government. Positioned as a pivotal force, the Chinese government orchestrates a unified and directive approach that is clearly reflected in the policy landscape exemplified as follows.

(9) ***各级财政**要加大对绿色低碳产业发展、技术研发等的**支持力度** … **(Fiscal support across all levels of government** should be **enhanced to accelerate** the development of green and low-carbon industries and technological R&D) …* (the State Council's Guideline on Achieving the Goals of Carbon Peak and Carbon Neutrality with Full and Faithful Implementation of the New Development Philosophy in 2021)

The Chinese policy discourse reflects a global vision through its active participation in 'international climate change discussions'. China's efforts to strengthen international exchanges and cooperation align with the Paris Agreement's fundamental principle that climate adaptation necessitates global collaboration. In particular, the 'Belt and Road' Initiative underscores China's commitment to partnering with numerous developing countries to 'accelerate the green transformation'. The following examples illustrate this.

(10) *积极参与应对**气候变化国际谈判** … (China will actively participate **in international climate change discussions**)* … (the State Council's Guideline on Achieving the Goals of Carbon Peak and Carbon Neutrality with Full and Faithful Implementation of the New Development Philosophy in 2021)

(11) *加快 **"一带一路"** 投资合作绿色转型, 支持共建 **"一带一路"** 国家开展清洁能源开发利用 … (We will **accelerate the green transformation** of **Belt and Road** investment and cooperation while supporting the development and deployment of clean energy in participating countries)* … (the State Council's Guideline on Achieving the Goals of Carbon Peak and Carbon Neutrality with Full and Faithful Implementation of the New Development Philosophy in 2021)

5.2.2. Stances of the American government on climate change

As presented in Table 5, American climate policy discourse encompasses a range of semantic domains, ordered by their LL values, reflecting diverse themes: W4 (Weather), W5 (Environmental Concerns), I3.2 (Professional Work and Employment), S7.1+ (Power Dynamics), G1.1 (Governmental Involvement), I2.2 (Business and Sales Strategies), S8+ (Assistance and Support), S5+ (Group Affiliation), O1.3 (Gaseous Substances and Materials) and T1.1.3 (Temporal Focus on the Future).

Table 5
Key Semantic Domains and Keywords in the American National Government Climate policies.

Rank	Tagset[a]	LL[b]	Semantic Domain	Keywords
1	W4	690.56	Weather	Climate
2	W5	684.98	Green issues	Environmental, environment
3	I3.2	679.8	Work and employment: Professionalism	Secretary
4	S7.1+	660.9	In power	Order, director, executive
5	G1.1	449.82	Government	Council, president
6	I2.2	411.68	Business: Selling	Agency
7	S8+	361.52	Helping	Advisor, support, guidance
8	S5+	326.5	Belonging to a group	Federal
9	O1.3	225.41	Substances and materials: Gas	Gas, air
10	T1.1.3	224.7	Time: Future	Shall, will

[a] Tagset is a collection of assignments that represent semantic categories.
[b] LL represents log-likelihood, with higher LL values indicating greater statistical significance.

The discourse of the U.S. government's climate policy is structured around two principal themes: 1. 'climate crisis and environmental concerns' and 2. 'responsibilities of specialised governmental agencies'. The first theme is reflected in dominant semantic domains, with W4 (Weather) and W5 (Environmental Concerns) directly linked to climatic and environmental issues. In conjunction with O1.3 (Gaseous Substances and Materials), these domains underscore the U.S. government's approach to addressing climate challenges and the strategies developed to manage them. Moreover, this narrative presents climate change as a critical issue with escalating risks and impacts across various sectors, reinforcing the urgency of action. Additionally, the term 'environmental' frequently appears in keyword concordances linked to environmental justice, suggesting that U.S. climate policy incorporates this concept within a broader dialogue on equity and systemic responses to climate challenges. This reflects a deep-seated engagement with issues of race, class and systemic inequality (refer to Examples 12–13).

(12) *The Federal Government faces broad exposure to the mounting risks and costs already posed by the **climate crisis*** (Catalyzing Clean Energy Industries and Jobs through Federal Sustainability in 2021).

(13) *It is critical that the Federal Government incorporate **environmental justice** considerations into sustainability and climate adaptation programs, and operations* (Catalyzing Clean Energy Industries and Jobs through Federal Sustainability in 2021).

The second theme, 'responsibilities of specialised governmental agencies', outlines a complex network of roles and structures that support the U.S. climate governance framework. The semantic domains of I3.2 (Professional Work and Employment), G1.1 (Government) and S5+ (Group Affiliation) converge around the central theme of governmental agency involvement. Notably, the term 'secretary' frequently appears in this domain, prominently featured in titles such as Secretary of Energy, Secretary of Commerce and Secretary of the Treasury. Furthermore, the 'Government' domain is characterised by the keyword 'council', emphasising the significance of administrative bodies such as the Council on

Environmental Quality and the Environmental Justice Advisory Council. Although these entities operate within the broader U.S. Federal Government, they play a crucial role in shaping environmental policy and governance (refer to Examples 14–15).

(14) *In coordination with the Chair of CEQ and the Director of OMB, the* ***Secretary of Energy*** *shall provide tools and technical support to agencies to develop targets for greenhouse gas emissions* (Catalyzing Clean Energy Industries and Jobs Through Federal Sustainability in 2021).

(15) *The Chair of the* ***Council on Environmental Quality*** *shall consider additional administrative steps and guidance to assist the Federal Acquisition Regulatory Council in developing regulatory amendments to promote increased contractor attention on reduced carbon emission and Federal sustainability* (Tackling the Climate Crisis at Home and Abroad in 2021).

Notably, these institutions appear more frequently than the broader entity of the 'Federal Government', which ranks only eighth in association with the keyword 'federal government' within the S5+ (Group Affiliation) domain. This ranking disparity suggests a strategic emphasis within the discourse, prioritising specific roles and agencies over the federal entity as a whole. The disparity implies that the discourse strategically highlights actionable and specialised governmental bodies rather than the federal structure in its entirety. Various agencies have the authority to implement, debate and even suspend policies enacted by others. Example 16 reveals how a specialised government agency, represented by 'The Secretary of the Interior', could direct law enforcement to halt certain activities of the American federal government.

(16) ***The Secretary of the Interior*** *shall, as appropriate and consistent with applicable law, place a* ***temporary moratorium*** *on all activities of the* ***Federal Government*** *relating to the implementation of the Coastal Plain Oil and Gas Leasing Program…* (Protecting Public Health and the Environment and Restoring Science to Tackle the Climate Crisis in 2021).

5.3. Stances of petroleum corporations on the Paris Agreement

5.3.1. Stances of Chinese Petroleum corporations on the Paris Agreement

China's three major petroleum companies—CNPC, CNOOC and Sinopec—engage with the Paris Agreement through two primary themes, namely, 'support for the agreement' and 'international cooperation'. This thematic focus precisely aligns with the general theme of 'engagement' identified in the earlier analysis of the Paris Agreement.

5.3.1.1. Support for the Paris Agreement. All three Chinese petroleum corporations have expressed full support for the Paris Agreement and a commitment to aligning their operations with its directives. CNPC, as shown in Example (17), demonstrates its support by striving to meet the agreement's specific goals.

(17) *In response to the* ***Paris Agreement*** *adopted at the Paris Climate Conference,* ***which aims to keep the global average temperature increase 'well below 2°C above preindustrial levels'*** *…, we are actively addressing climate change and advancing a green, low-carbon transition with the goal of* ***achieving near-zero emissions by around 2050*** (CNPC Climate Change Webpage).

In Example (18), CNOOC presents itself as a 'responsible energy corporation' that actively responds to the Paris Agreement.

(18) *As a responsible energy corporation, CNOOC has actively engaged with the* ***Paris Agreement*** *and taken proactive measures to address climate change* (CNOOC Environmental Protection Webpage).

While Sinopec does not explicitly reference the Paris Agreement, it highlights Carbon Capture, Utilisation and Storage (CCUS) as essential to achieving the agreement's objectives, as demonstrated in the following example.

(19) *According to the Intergovernmental Panel on Climate Change, the goals outlined in the* ***Paris Agreement*** *cannot be achieved without CCUS. Without CCUS, the cost of global carbon emission reduction would increase significantly* (2021 Sinopec Corporate Social Responsibility Report).

5.3.1.2. International cooperation. In addition to their commitment to the Paris Agreement, Chinese petroleum companies emphasise international cooperation. This thematic focus echoes the overarching principle of global engagement embedded within the Paris Agreement. CNPC underscores its role in global climate initiatives as follows:

(20) *CNPC actively engages in* ***international climate change cooperation****, collaborating with other member companies to leverage its expertise and contribute to efforts addressing climate change while advancing the green and low-carbon transition of the oil and gas industry* (2021 CNPC Corporate Social Responsibility Report).

Similarly, CNOOC actively fosters international cooperation through the following projects:

(21) *As an international energy company, CNOOC Limited has established operations in over 20 countries or regions across 6 continents … Its* ***international cooperation projects*** *have created numerous job opportunities for local communities* (2021 CNOOC Environmental, Social and Governance Report).

Sinopec enhances its international engagement, particularly through its participation in the United Nations Global Compact, as demonstrated in the following:

(22) *Sinopec remains committed to strengthening* ***international cooperation*** *on climate change, including its ongoing participation in the United Nations Global Compact* (2021 Sinopec Corporate Social Responsibility Report).

5.3.2. Stances of American petroleum corporations on the Paris Agreement

In response to the Paris Agreement, the discourse of three American petroleum corporations reflects a delicate approach to balancing global climate issues with business operations. Thematic analysis of their discourses reveals two primary themes: 'commitment to aligning with the Paris Agreement's goals' and 'strategic focus on maintaining traditional energy production and financial interests'.

5.3.2.1. Commitment to aligning with the Paris Agreement's goals. The discourse of American petroleum corporations reflects a unified theme of support for the Paris Agreement's objectives. Chevron, as illustrated in Example (23), explicitly endorses the agreement's net-zero ambitions.

(23) *At Chevron, we believe the future of energy is lower carbon, and we support the global net zero ambitions of the* ***Paris Agreement*** (2021 Chevron climate change resilience report).

ExxonMobil and Marathon Oil, as demonstrated in Examples (24) and (25), respectively, affirm their commitment to the goals of the Paris Agreement. ExxonMobil acknowledges the significance of the agreement and conveys its alignment with its objectives, while Marathon Oil emphasises efforts to reduce GHG and methane emissions intensity in line with the agreement's goals.

(24) *We're committed to* ***continuing to*** *provide these critical products, working toward the goals of the* ***Paris Agreement****, and creating value for all stakeholders* (ExxonMobil Advancing Climate Solutions 2022 Progress Report).
(25) *In 2022, we made reductions in greenhouse gas (GHG) and methane emissions intensity, consistent with the* ***Paris Climate Agreement*** *objectives* (2022 Marathon Sustainability Report).

5.3.2.2. Strategic focus on maintaining traditional energy production and financial interests. Despite expressing support for the Paris Agreement, American petroleum corporations demonstrate a clear commitment to sustaining traditional energy production. Chevron, as illustrated in Example (26), affirms its continued oil production, justifying its position by emphasising petroleum's necessity and indispensability in society. Similarly, ExxonMobil, in Example (27), asserts that the Paris Agreement does not mandate fossil fuel production reductions for individual companies. Furthermore, ExxonMobil echoes Chevron's perspective that emissions are primarily driven by societal demand, shifting responsibility from the supplier to society.

(26) *Chevron supports the Paris Agreement and is committed to addressing climate change* ***while continuing to deliver energy that supports society*** (Chevron supports well-designed climate policy webpage).
(27) *The Paris Agreement does not contemplate or require individual companies to decrease production to align with the goal of maintaining global temperature rise to below 2°C.* ***The structure of the agreement recognizes that energy-related emissions are driven by society's demand for energy – not its supply*** (ExxonMobil 2021 Energy & Carbon Summary).

Marathon Oil's discourse, as demonstrated in Example (28), emphasises risk management and regulatory adaptation rather than environmental leadership. The company's preparations for various regulatory scenarios suggest a primary focus on mitigating negative impacts on its business rather than driving environmental innovation. Notably, it expresses significant concern about potential business disruptions and harm to shareholder interests following the U.S. rejoining the Paris Agreement during President Joe Biden's tenure.

(28) *Although uncertain, these developments (**rejoining the Paris Agreement**) could: set limits on emitted greenhouse gases; reduce the demand for crude oil, … Marathon Oil monitors and **prepares for regulatory changes that could impact our business** and has implemented mitigation practices that help ensure resilience to a number of regulatory and legislative scenarios* (Marathon Oil Climate Change Webpage).

5.4. Stances of petroleum corporations on national policies

5.4.1. Stances of Chinese Petroleum corporations on national policies

The climate discourse of three Chinese petroleum corporations converges on two key themes: 'transition to sustainable energy management' and 'a policy-driven climate strategy'. These two themes closely align with the Chinese government's emphasis on comprehensive energy management and the direction of national policy.

5.4.1.1. Transition to sustainable energy management. The discourses of Sinopec, CNOOC and CNPC consistently emphasise sustainable energy consumption management, the implementation of energy efficiency initiatives and the transition to low-carbon energy sources, underscoring the overarching theme of 'transition to sustainable energy management'.

Sinopec's climate communication highlights the need to 'enhance the regulation of overall energy consumption and intensity', emphasising the managerial role of the entire system. Its explicit reference to 'the control of fossil energy use' demonstrates a clear strategic shift towards reducing carbon emissions (see Example 29).

(29) *We should enhance the regulation of overall energy consumption and intensity, prioritising the control of fossil energy use, to transition towards a system that limits total carbon emissions and intensity* (2022 Sinopec Corporate Social Responsibility Report).

CNOOC, as demonstrated in Example (30), considers 'energy conservation and emission reduction' to be crucial mechanisms for transforming development patterns and enhancing competitiveness, reflecting a strategic management approach.

(30) *As an advocate and practitioner of a low-carbon economy, CNOOC consistently upholds a green, low-carbon and circular development philosophy, viewing **energy conservation and emission reduction** as vital strategies for transforming its development model and strengthening core competitiveness* (CNOOC Climate Change Webpage).

CNPC's contributions, as highlighted in Example (31), centre on the implementation of a green action plan that advances energy conservation, emission reduction and the integration of clean energy into production processes. It shares a similar perspective with CNOOC, stressing the importance of reducing fossil energy consumption.

(31) *We advance **the green action plan by implementing energy conservation, emission reduction and clean energy integration** in production, aiming to **minimise fossil energy consumption** and CO_2 emissions at the source, ultimately achieving carbon reduction* (2022 CNPC Corporate Social Responsibility Report).

5.4.1.2. A policy-driven climate strategy. The three Chinese petroleum corporations explicitly align their strategies with the overarching environmental objectives set by the Chinese government, emphasising proactive responses to government policies.

CNPC's dedication to achieving the carbon peak and carbon neutrality targets set by the central government reflects its proactive approach to climate change. This stance underscores CNPC's alignment with national climate objectives and its role as an active participant in the national climate campaign, exemplified as follows.

(32) *In alignment with the Paris Agreement, we (CNPC) have proactively engaged in international climate governance and actively addressed climate change by implementing the **carbon peak and carbon neutrality targets set by the Chinese government*** (2022 CNPC Corporate Social Responsibility Report).

Sinopec, as outlined in Example (33), explicitly affirms its commitment to implementing the directives of the Communist Party of China (CPC) Central Committee and the State Council concerning carbon peak and carbon neutrality targets. Its dedication to achieving China's 'dual-carbon' goal underscores its support for national efforts towards a green and low-carbon transformation.

(33) *Sinopec **firmly implements the directives of the CPC Central Committee and the State** on carbon peak performance and carbon neutrality … contributing to the green and low-carbon transformation of China's energy and chemical industry and the **achievement of the dual-carbon goal*** (Sinopec Tackling Climate Change Webpage).

CNOOC's reference to President Xi Jinping's well-known quote 'Clear waters and lush mountains are invaluable assets (绿水青山也是金山银山)', as illustrated in Example (34), aligns the company with a broader philosophical and political framework that emphasises environmental preservation as essential to sustainable development. This quote embodies a core principle of ecological civilisation—a concept central to China's environmental policy under President Xi's leadership—and is closely linked to the traditional Chinese *Daoist* philosophy of harmony between humans and nature.

(34) *CNOOC Limited has consistently upheld the principle that* ***clear waters and lush mountains are invaluable assets****, actively addressing climate change and implementing the* ***national targets of achieving carbon peaking by 2030 and carbon neutrality by 2060*** (2021 CNOOC Corporate Social Responsibility Report).

5.4.2. Stances of American petroleum corporations on national policies

Compared to their Chinese counterparts, American petroleum corporations show a weaker correlation in aligning with national government policies. However, these corporations demonstrate a stronger tendency to influence policy decisions. Their response to U.S. government policies can be primarily categorised into two themes, namely, 'selective policy support and active lobbying' and 'advocacy for a market-driven carbon pricing policy'.

5.4.2.1. Selective policy support and active lobbying. Regarding policy support, the three American petroleum corporations—Chevron, ExxonMobil and Marathon Oil—demonstrate a stance that can be described as cautiously supportive of climate policies focused on emission reduction. Their support is discernibly selective and not consistently applied across all national policies.

Chevron's discourse, as illustrated in Example (35), emphasises support for 'balanced and measured' policies while cautioning against 'duplicative and inefficient regulations'. This stance suggests a preference for well-designed policies, implicitly critiquing certain regulations as inefficient and reflecting a mildly critical perspective on government policies.

(35) *Chevron supports* ***balanced and measured policy****: Involve all sectors to maximize efficient and cost-effective reductions while allocating costs equitably, gradually, and predictably;* ***avoid duplicative and inefficient regulations****; balance economic, environmental, and energy needs* (Chevron Supports Well-designed Climate Policy Webpage).

Marathon Oil's discourse, as reflected in Example (36), conveys a similar perspective, outlining its definition of a successful public policy. The primary criterion is the protection of its vested interests, particularly ensuring that oil and gas production remains unaffected. This criterion underscores the importance of safeguarding business interests while simultaneously addressing broader environmental concerns.

(36) *We believe that* ***successful public policy*** *must recognize that oil and gas underpin our standard of living and American oil and gas is critical to our national security and economic prosperity. We will* ***support policy that****: Requires proportional participation from all sectors of the economy; Balances economic, environmental and energy security needs* (Marathon Oil Political Contributions & Lobbying Webpage).

ExxonMobil's discourse, as presented in Example (37), similarly conditions its policy support, emphasising stability and long-term business viability. This stance suggests selective endorsement of policies that safeguard its operational continuity and profitability amid the evolving energy landscape.

(37) *ExxonMobil* ***supports policies that*** *promote a stable investment climate for long-term business viability and continues to communicate our positions to policy stakeholders and governments around the world* (ExxonMobil Advancing Climate Solutions Webpage).

Concurrently, these American corporations actively engage in lobbying efforts to shape policy decisions, particularly when they believe certain regulations or proposals could negatively affect their business operations.

Chevron, as illustrated in Example (38), presents its lobbying activities as efforts to advance policy measures that align with its goal of providing 'affordable, reliable and ever-cleaner energy'. The company's engagement in policy discussions on the global energy transition reflects a strategic approach to influencing regulations in a manner that supports its corporate interests.

(38) *Our* ***climate lobbying activities*** *seek to encourage appropriate measures supporting our ambition to deliver affordable, reliable, and ever-cleaner energy, including on issues of public policy like the global energy transition* (2021 Chevron Climate Change Resilience Report).

Marathon Oil, as demonstrated in Example (39), conveys its policy lobbying through financial contributions to candidates and organisations that advocate for responsible energy development. The company's explicit mention of 'direct corporate contributions' underscores its commitment to supporting policy frameworks that align with its positions and interests.

(39) *We **contribute to candidates and organizations** that support responsible energy development. In 2020, Marathon Oil made **direct corporate contributions** totaling $62,500* (Marathon Oil Political Contributions & Lobbying Webpage).

ExxonMobil demonstrates a multifaceted approach to policy engagement, as illustrated in Example (40). ExxonMobil explicitly highlights direct policy engagement, collaboration with trade associations and participation in industry initiatives such as the Oil & Gas Climate Initiative. This approach underscores the corporation's commitment to shaping policies through diverse channels, reflecting a comprehensive strategy for influencing energy and climate policies.

(40) *ExxonMobil **engages on policy directly,** through trade associations and focused industry collaboration efforts, such as the Oil & Gas* Climate Initiative (ExxonMobil 2021 Energy & Carbon Summary).

5.4.2.2. Advocacy for a market-driven carbon pricing policy. In the discourse presented by American petroleum corporations on their lobbying of national policies, a recurring theme is their advocacy for a market-driven carbon pricing policy.

Across these companies, there is a shared consensus that carbon pricing is the most effective mechanism for leveraging market forces to reduce emissions. This perspective is evident in Chevron's assertion in Example (41), which emphasises that carbon pricing should be central to policy efforts aimed at lowering GHG emissions. ExxonMobil reinforces this stance in Example (42), advocating for 'market-based approaches' that include an 'economy-wide price on carbon'. Similarly, Marathon Oil's viewpoint in Example (43) aligns with this market-driven strategy, underscoring the belief that 'an economy-wide, market-based policy' imposing a uniform cost on carbon is the most efficient way to drive emission reductions.

(41) *We believe a price on carbon is the most efficient mechanism to **harness market forces** to reduce emissions. **Carbon pricing** should be the primary policy tool to achieve greenhouse gas emissions reduction goals* (Chevron Supports Well-designed Climate Policy Webpage).
(42) *The Company supports **market-based approaches** to reduce emissions, including further cost-effective regulation of methane and an **economy-wide price on carbon*** (ExxonMobil 2021 Energy & Carbon Summary).
(43) *We believe **an economy-wide market-based policy** that places a uniform, predictable **cost on carbon** will drive emission reductions at the lowest cost to society* (2021 Marathon Climate Report).

Notably, in the discourse of American petroleum companies regarding market-driven carbon pricing, there is an apparent absence of direct engagement with the central theme of 'environmental justice' as emphasised in U.S. federal policies. This omission indicates that although these companies prioritise economic and market efficiency in their carbon pricing strategies, incorporating justice considerations into their policy advocacy does not appear to be a primary focus in their current climate approach.

6. Discussion and conclusions

6.1. Discussion: multiple influences in shaping corporate climate discourse

This study analysed how American and Chinese petroleum corporations articulate their climate positions in relation to two key shareholders—the Paris Agreement and their respective national governments—and explored various influences shaping corporate climate discourse.

The findings indicate that the Paris Agreement underscores global engagement, developmental capacity and formal, obligatory commitments to directives. The Chinese policy discourse presents the government as a central orchestrator, actively guiding national policy within a global framework. Meanwhile, U.S. policy discourse focuses on addressing immediate climate crises and environmental justice while highlighting the roles of various agencies. In response to these two key stakeholders, Chinese corporations' discourse aligns closely with the Paris Agreement, centring around two themes: 'transition to sustainable energy management' and 'policy-driven climate strategy'. However, American corporations adopt a more nuanced approach, balancing adherence to the global agreement with business operations through 'selective policy support and active lobbying' and 'advocacy for a market-driven carbon pricing'.

The differences in how Chinese and American petroleum corporations position themselves concerning two key stakeholders stem primarily from three factors. First, from a corporate ownership perspective, major American petroleum corporations are privately owned, with ownership exclusively being with shareholders. This structure results in a strong emphasis on shareholder interests in their discourse. Consequently, the American federal government, as a policymaker, has limited influence in compelling these business entities to align with its directives. American corporations often perceive climate policies and regulations as risks to be managed. However, leading Chinese petroleum corporations operate as state-owned enterprises, with the state holding ownership or control over their capital. These enterprises are typically established to achieve specific economic regulatory goals, ensuring the harmonisation of various aspects of the national economy (Li &

Zhao, 2018). Therefore, they are required to consistently adhere to government policies, leading to a significantly stronger regulatory influence over corporate practices.

Second, from the perspective of the corporate–government relationship, corporate lobbying in the U.S. is common and constitutionally protected. Moreover, it falls under the First Amendment, guaranteeing freedom of speech and the right to petition the government for redress of grievances (Meiklejohn, 1961). American corporations often view lobbying as a proactive strategy for managing regulatory risks (Favotto & Kollman, 2021). In contrast, lobbying is rarely mentioned in the discourse of Chinese corporations. Instead, they primarily align with policy directions set by the central government, focusing on compliance and relationship-building with government authorities.

Third, cultural traditions also play a role in shaping corporate behaviour. Chinese culture is characterised by high power distance, where individuals and organisations typically respect and trust authority (Hofstede, 2011). In this study, the Chinese corporations' emphasis on 'transition to sustainable energy management' and 'a policy-driven climate strategy' reflects this cultural tradition of respecting authority. In contrast, American culture is shaped by a tradition of low power distance, emphasising individualism, self-reliance and corporate independence (Hofstede, 2011; Potter, 2019). In this study, American corporations' 'selective policy support and active lobbying' and 'advocacy for a market-driven carbon pricing' embodies this decentralised cultural tradition.

6.2. Academic and practical implications

This study addresses gaps in existing research by integrating stakeholder theory with corporate climate discourse analysis to examine how Chinese and U.S. petroleum corporations position themselves in terms of two critical stakeholders—the Paris Agreement and their respective national governments. It is observed that corporate climate discourse has been extensively addressed, primarily focusing on self-reporting and self-representation, especially self-legitimisation and self-promotional discourse and strategies (e.g., Bhatia, 2012; Dahl, 2022; Jaworska, 2018; Megura & Gunderson, 2022; Wang & Liu, 2024). Nevertheless, its full engagement with the key stakeholders is underexplored. We argue that adopting a comparative and dynamic approach contributes positively to a full understanding of the whole communicative event. By providing this novel perspective and integrating stakeholder theory with corporate climate discourse, this study highlights how corporations address and navigate diverse stakeholder expectations, balance competing priorities and examine the various influences shaping corporate climate discourse.

Methodologically speaking, this study demonstrates the critical role of discourse in analysing corporate stances and how nuanced linguistic analysis could contribute to the widely-discussed topic of climate change and global governance. When tackling different types of data, we applied the mixed-methods design by integrating quantitative and qualitative thematic analyses. It presents a comprehensive and dynamic overview of how American and Chinese petroleum corporations construct their climate discourses, particularly in relation to the influential Paris Agreement and national government policies. Thus this study exhibits the viability and vitality of incorporating different discourse analytic tools.

This study further provides practical insights to climate policy makers by enhancing their understanding of how the two biggest carbon emitters—China and U.S.—discursively frame these policies and what are the petroleum corporations' climate positions in the two nations. As climate change has become a pressing global issue, it is of critical importance to formulate well-informed policies, enhance mutual understanding and negotiation and foster international collaboration. Furthermore, this study could serve as a valuable reference for corporate communication students and practitioners, aiding them in tailoring messaging strategies for diverse stakeholders while considering the various factors that shape corporate climate discourse.

6.3. Limitations and suggestions for future research

The study acknowledges two limitations and offers suggestions for future research. First, this study relies on data from six major petroleum corporations in China and the U.S., which may not fully capture the discourse of other entities in these countries. Expanding the sample to include a more diverse range of petroleum companies—from leading multinational corporations to local and regional players—would enhance the breadth and representativeness of the findings. Second, the data on national government policies in the U.S. and China were drawn from the period when the Paris Agreement was adopted and continuously enforced in each country to ensure comparability. Therefore, due to his administration's opposition to the Paris Agreement and formal withdrawal from the global climate pact, policies from President Donald Trump's tenure were excluded from the analysis. A comparative discourse study of climate policies under the administrations of President Donald Trump and President Joe Biden could offer in-depth insights into achieving a comprehensive understanding of U.S. government stances on climate change.

Declaration of competing interest

None.

CRediT authorship contribution statement

Jieyun Feng: Conceptualization, Writing – review & editing, Supervision, Writing – original draft, Methodology, Investigation, Formal analysis, Validation. **Wenze Lu:** Conceptualization, Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Validation. **Ming Liu:** Conceptualization, Methodology. **Wenqing Yu:** Writing – original draft, Data curation.

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Data availability

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Jieyun Feng is professor at the Department of Business English, School of International Studies, University of International Business and Economics, Beijing, China. Her research interests include business discourse, intercultural communication, discourse on cultural heritage and tourism, AI-assisted English learning.

Wenze Lu is postdoctoral fellow at the School of Nursing, Faculty of Health and Social Sciences, The Hong Kong Polytechnic University, HKSAR, China. His research interests include media communication, intercultural discourse analysis, psychosocial intervention, medical humanities and healthcare linguistics.

Ming Liu is assistant professor at the Department of Chinese and Bilingual Studies, Faculty of Humanities, The Hong Kong Polytechnic University, HKSAR, China. His research interests include corpus linguistics, critical discourse analysis and intercultural communication.

Wenqing Yu is general secretary of the General Office, WeBank Co., Ltd., Shenzhen, China. She holds a MA degree from the University of International Business and Economics, Beijing, China, and a MSc degree from the University of Birmingham, UK.